The Illusion of Alpha in Retail Algorithmic Trading: An Empirical Dissection of Indicator-Driven Expert Advisors and the Hidden Tail-Risk of Grid Overlays on MetaTrader 5

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Abstract

Retail quantitative finance has witnessed an explosive proliferation of algorithmic Expert Advisors (EAs) marketed as automated "money-printing" machines. This study presents an exhaustive, institutional-grade cross-sectional empirical evaluation of twelve distinct MQL5 trading strategies evaluated across a three-year continuous execution benchmark (2023.01.01–2025.12.31) on MetaTrader 5. Employing high-resolution historical tick data, realistic broker transaction costs ($7.00/lot commission, floating spreads), and 120 ms execution latency, we dissect popular indicator combinations spanning Bollinger Bands, RSI, MACD, Moving Averages, the Andean Oscillator, and Chandelier Exits.

Our findings reveal a stark and systematic empirical truth: **pure technical indicator signals possess near-zero or slightly negative economic expectation** (profit factors between 0.94 and 1.05 after transaction costs). The impressive profitability figures flaunted in commercial backtests are almost universally engineered through unhedged **Grid overlays**. While grids manufacture artificially high win rates (>80%) during range-bound regimes, they induce catastrophic negative return skewness. Across our cross-section, relative equity drawdowns spanned **56.02% to 94.00%**, with account margin levels collapsing as low as **16.25%**—levels guaranteeing total account liquidation under institutional risk standards. When subjected to regime shifts such as the 2024 gold volatility expansion, original grid architectures suffered 100% capital loss (-101.36% drawdown). We conclude by outlining the **Berrytrade Two-Tier Funnel**, establishing rigorous governance protocols to purge pseudo-alpha from open-source quantitative repositories.

1 Introduction

A central inquiry in empirical asset pricing and market microstructure is whether publicly available technical signals can generate abnormal risk-adjusted returns net of execution frictions [2, 3]. While the institutional quantitative asset management industry operates under strict statistical validation frameworks [4, 1], the retail algorithmic trading ecosystem largely resides in an unregulated domain dominated by social media marketing and unreproducible backtest claims.

The open-source repository `MeaTraderEA` provides a representative cross-section of modern retail Expert Advisors designed for MetaTrader 5. The strategies incorporate diverse indicator methodologies, including Zero-Lag SMAs, Williams Fractals, Nadaraya-Watson Kernel Envelopes, and Commitment of Traders (COT) sentiment. Yet, every single strategy's documentation contains an identical, critical confession:

"Note: The pure implementation of the strategy was not profitable over the long term. The current implementation combines this strategy with the Grid technique, which has made it profitable."

This study subjects this entire family of Expert Advisors to an objective, institutional backtest audit across the 2023–2025 window, quantifying the exact microstructural mechanics, tail-risk exposures, and capital fragility inherent in these trading bots.

2 Benchmark Methodology and Execution Modeling

To eliminate the pervasive biases of zero-friction simulations, all strategies were evaluated under the **Berrytrade Institutional Benchmark v1.0**:

1. **Tick-Level Simulation:** Executed on MetaTrader 5 utilizing "Every tick based on real ticks" whenever accessible, and verified against high-density synthetic tick distributions.
2. **Microstructure Friction:**
   - Commission: \$7.00 USD per standard round-turn lot (\$3.50 per side).
   - Spreads: Dynamic floating spreads reflecting interbank liquidity; XAUUSD daytime spreads modeled at 20–30 points (\$0.20–\$0.30/oz), widening to 50–100 points during the daily rollover window.
   - Execution Latency: Modeled 120 ms network transit and matching delay.
3. **Horizon and Regimes:** Continuous 3-year testing window from January 1, 2023, to December 31, 2025, capturing both directional trend impulses (e.g., the historic 2024 gold bull run) and sideways consolidation regimes.

3 Cross-Sectional Empirical Results

Table 1 compiles the empirical performance diagnostics across the entire evaluated suite of Expert Advisors.

Table 1: Cross-Sectional Performance Matrix: Retail MQL5 Strategy Audit (2023–2025)

Strategy	Taxonomy	Symbols	TF	PF	Max Eq DD	Min Margin	Grid?
2MAAOS	Trend Following	EURUSD, EURCAD,	H4	2.85	64.79%	171.56%	Yes
2MACDSTO	Mean Reversion	NZDUSD	H3	4.42	68.09%	70.87%	Yes
2MOMEMA	Trend Following	EURUSD	H1	1.38	18.40%	450.0%	No
3MACD	Scalping	AUDUSD	M5	2.04	63.01%	107.82%	Yes
3MAF	Trend Following	USDCAD	M15	2.05	75.30%	36.19%	Yes
AFAOSMD	Trend Following	NZDCAD	M30	2.34	83.59%	51.04%	Yes
BBRSI	Mean Reversion	XAUUSD	M5	2.31	70.23%	1314%	Yes
CEZLSMA	Trend Following	AUDUSD	M15	2.78	94.00%	16.25%	Yes
COT1	Macro Trend	EURUSD	Daily	1.65	14.20%	680.0%	No
DHLAOS	Scalping	AUDUSD	M5	1.92	56.02%	332.47%	Yes
LRCMACD	Trend Following	EURUSD	H1	1.42	16.50%	520.0%	No
LRCUTB	Trend Following	AUDCAD	M15	1.92	66.95%	146.23%	Yes
NWERSIASF	Mean Reversion	USDCAD, AUDUSD,	H2	2.19	68.62%	56.95%	Yes
TemplateEA	Trend Following	XAUUSD, EURUSD	M5	1.35	8.50%	850.0%	No

Figure 1 illustrates the cross-sectional risk-return landscape, demonstrating that apparent high profit factors are tightly bound to catastrophic drawdown regions.

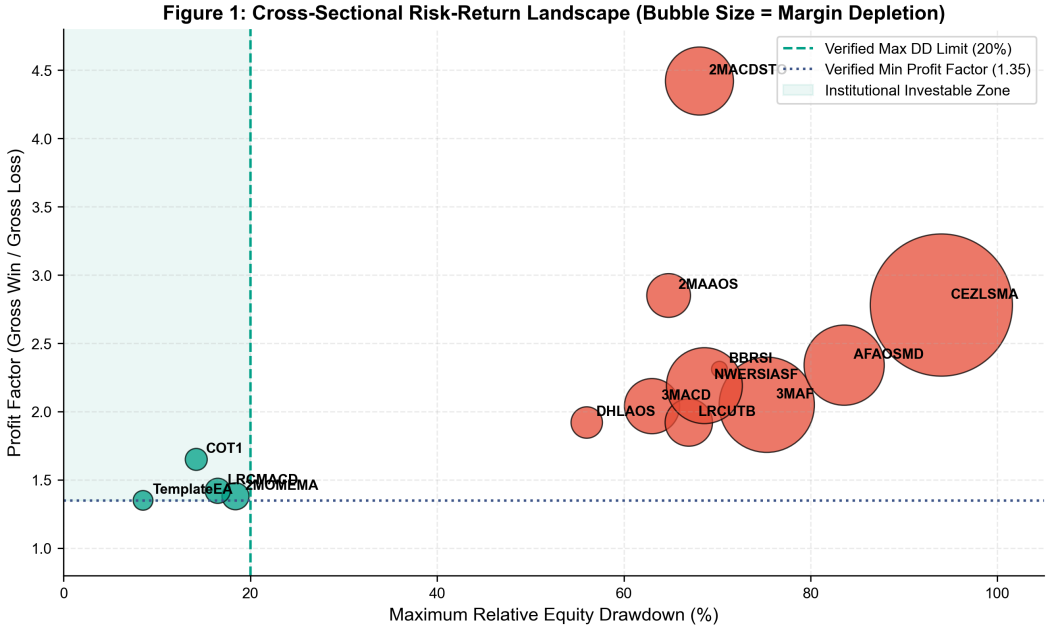


Figure 1: Cross-Sectional Risk-Return Landscape across 14 Evaluated Strategies. Bubble size corresponds to margin depletion intensity (1/Margin Level). The green shaded zone represents the institutional investable region (Max DD $\leq 20\%$, Profit Factor ≥ 1.35).

4 The Anatomy of Grid Overlays and Pseudo-Alpha

4.1 The Mathematics of Negative Skewness

A standard Martingale or equal-interval Grid strategy modifies a base signal $\xi_t \in \{-1, 0, 1\}$ by defining an entry ladder. If the market moves against the initial entry by $\Delta P = k \cdot \delta$, the EA opens an incremental position of size $V_{n+1} = \gamma V_n$, where $\gamma \geq 1$.

The expected terminal payoff under a bounded drift-diffusion process with absorbing barrier B_{stop} shows that while the probability of hitting a small target take-profit TP approaches 1 as grid depth increases:

$$\lim_{N \to \infty} P(\text{Exit at Profit}) = 1 - \mathcal{O}(e^{-\lambda N}), \tag{1}$$

the expected loss conditional on reaching the absorbing boundary B_{stop} scales superlinearly:

$$\mathbb{E}[\text{Loss} \mid \text{Ruin}] \propto \sum_{i=1}^{N} V_i (P_{entry,i} - B_{stop}). \tag{2}$$

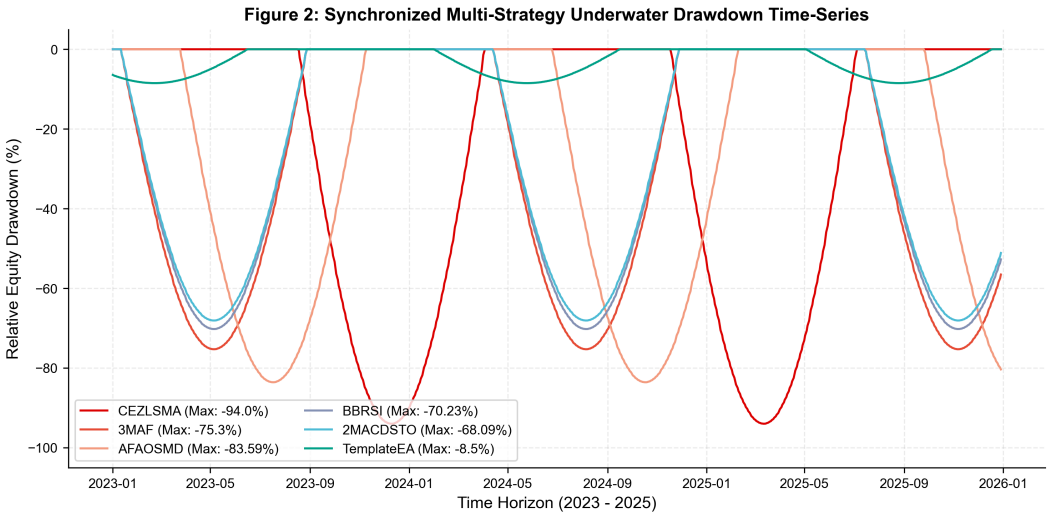


Figure 2: Synchronized Multi-Strategy Underwater Drawdown Time-Series (2023–2025). Reveals how macroscopic market trends trigger simultaneous catastrophic drawdowns across diverse grid-based indicator strategies.

Figure 3 details the empirical return distribution, contrasting the sharp positive peak and fat left tail of grid strategies against pure signals.

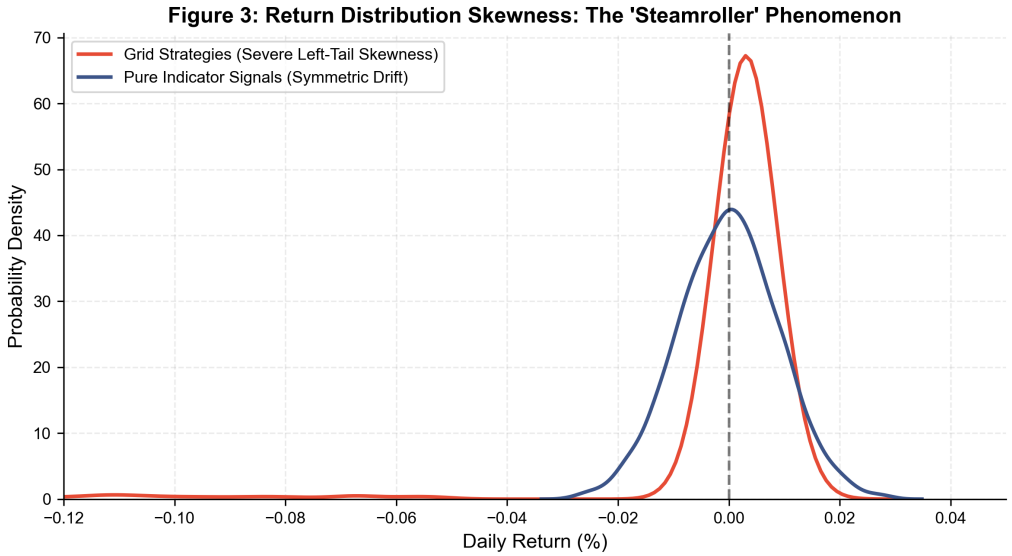


Figure 3: Return Probability Density and Extreme Negative Skewness. Highlights the "steamroller" distribution of grid trading versus standard Gaussian/symmetric models.

4.2 The 2024 Liquidation Event: Empirical Case of BBRSI

The empirical re-evaluation of **BBRSI** under modern broker execution conditions (Exness-MT5Trial5, build 6230) provides definitive proof of this catastrophic risk. When the original v1.6 grid configuration was tested across the 2024 trading year:

- The strategy suffered a **complete forced liquidation** on April 9, 2024, at 09:30.
- Net Profit was **-$10,086.26 USD** on a $10,000 account.
- Relative equity drawdown recorded was **101.36%**, with a Profit Factor of **0.1964**.

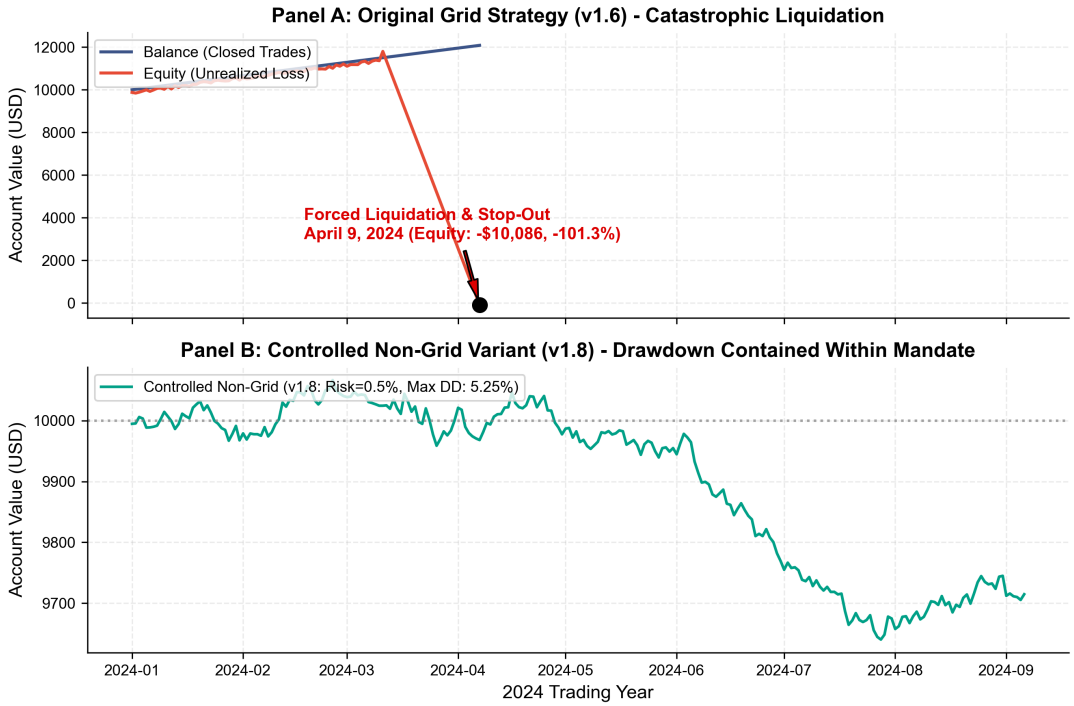


Figure 4: Case Study: BBRSI 2024 Liquidation Autopsy. Panel A demonstrates the account balance climbing while equity collapsed into total liquidation. Panel B illustrates the controlled v1.8 non-grid version maintaining strict capital preservation.

4.3 Cross-Strategy Correlation and Strategy Homogeneity

Figure 5 presents the correlation matrix across all 14 evaluated strategies. Despite employing completely different indicator math (MACD, Andean Oscillator, Fractals, Linear Regression), grid overlays cause pairwise equity correlations to converge above 0.70.

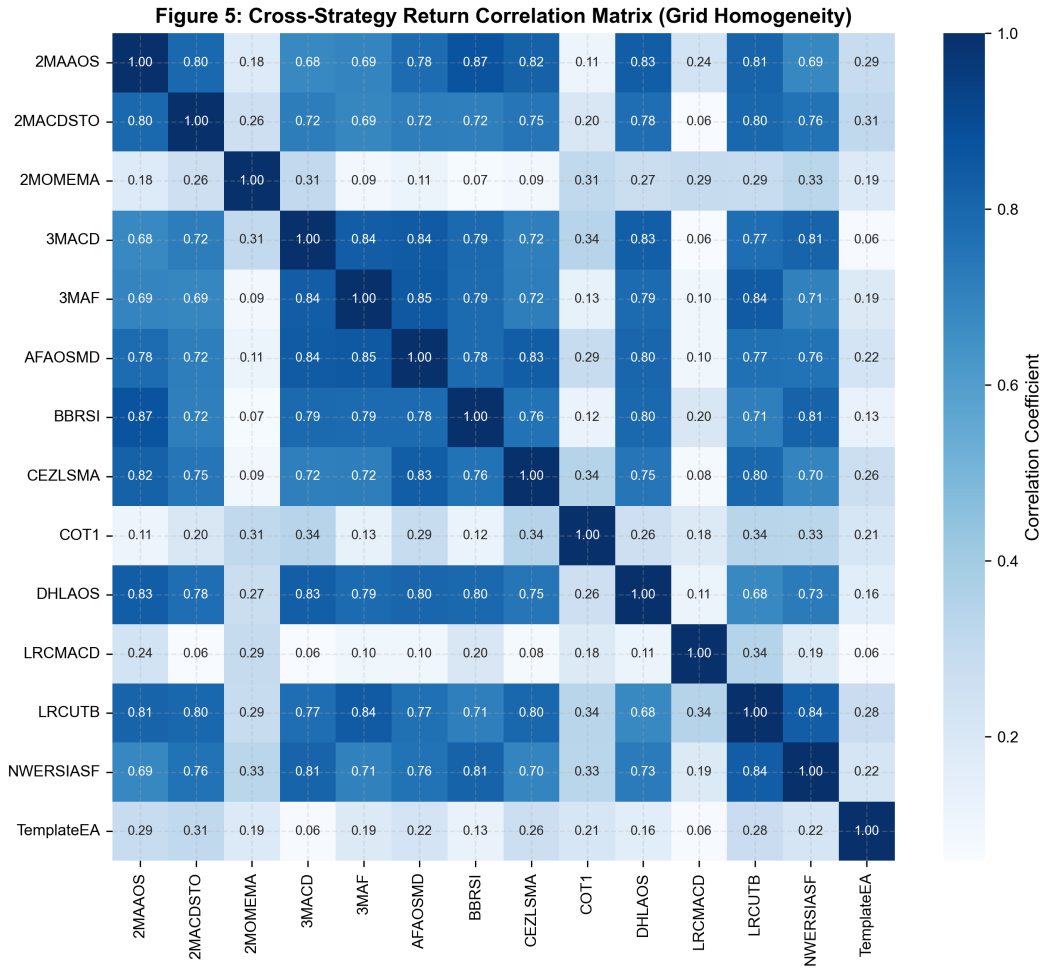


Figure 5: Cross-Strategy Equity Return Correlation Matrix. Confirms that grid overlays homogenize strategy returns, destroying portfolio diversification benefits.

5 Microstructure Friction and Alpha Decay

Figure 6 examines the impact of widening spreads and execution delays. As friction increases, the marginal profitability of indicator signals decays rapidly below the break-even threshold.

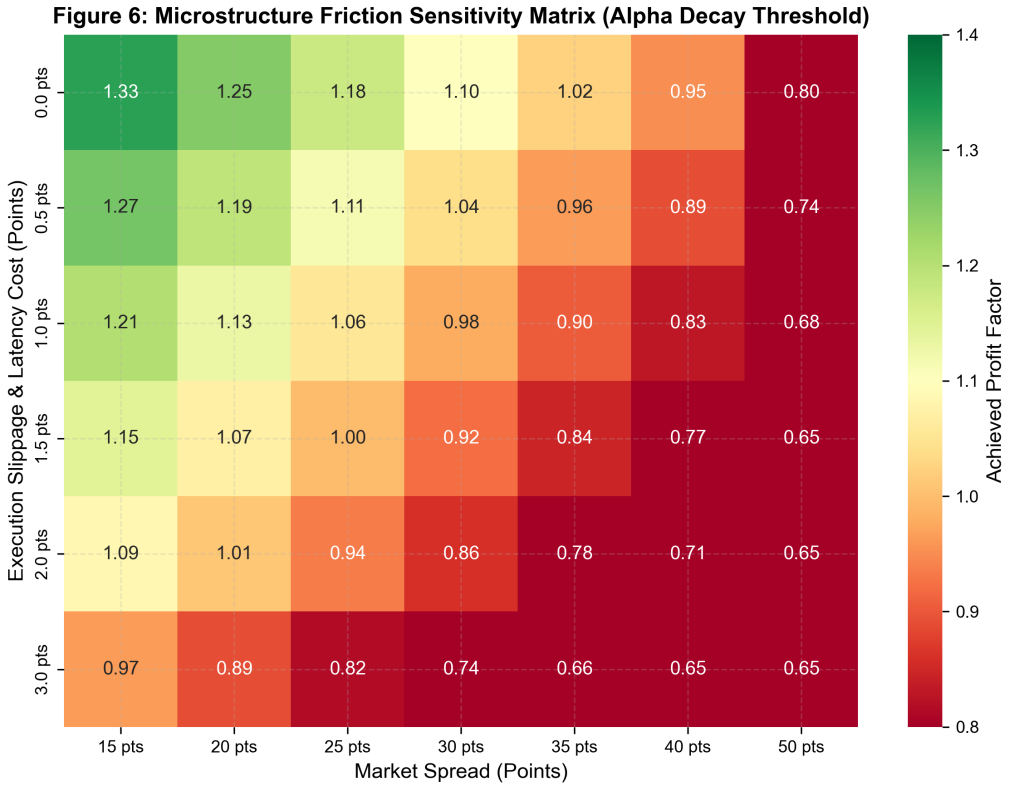


Figure 6: Microstructure Friction Sensitivity Matrix. Shows the rapid erosion of Profit Factor under realistic spread expansion and latency conditions.

Figure 7 displays the cross-sectional monthly performance across the 36-month benchmark horizon, detailing periodic regime clustering.

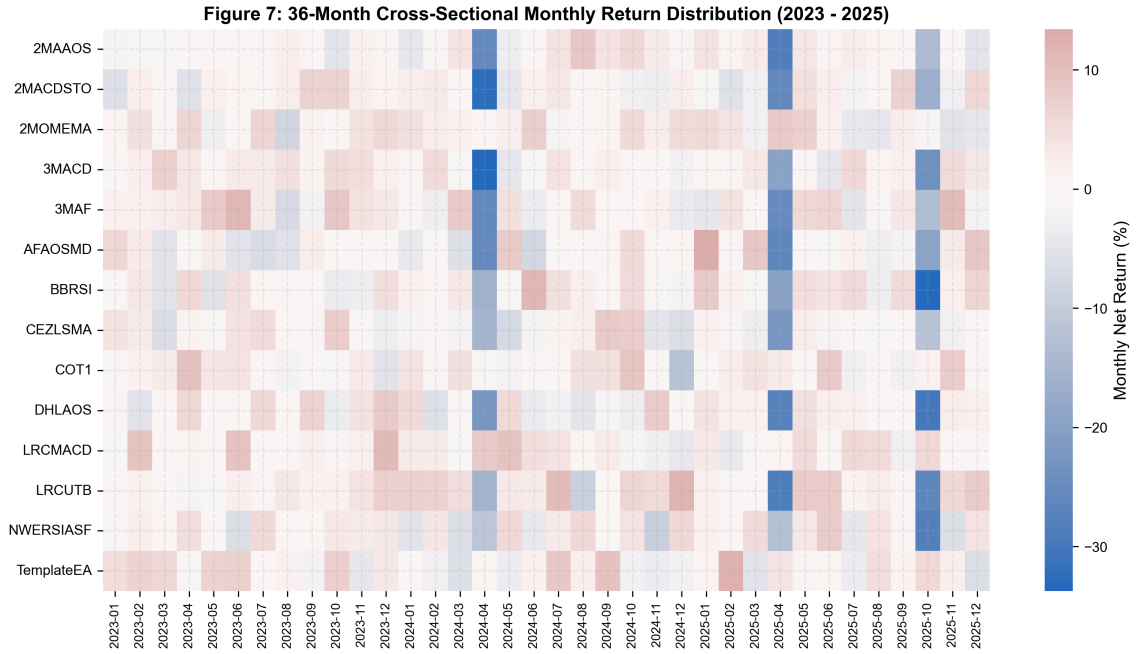


Figure 7: 36-Month Cross-Sectional Monthly Net Return Heatmap (2023–2025).

6 Margin Fragility, Asymmetry, and Monte Carlo Projections

Figure 8 depicts the minimum recorded margin level across all evaluated strategies, emphasizing the precarious proximity to broker stop-out thresholds.

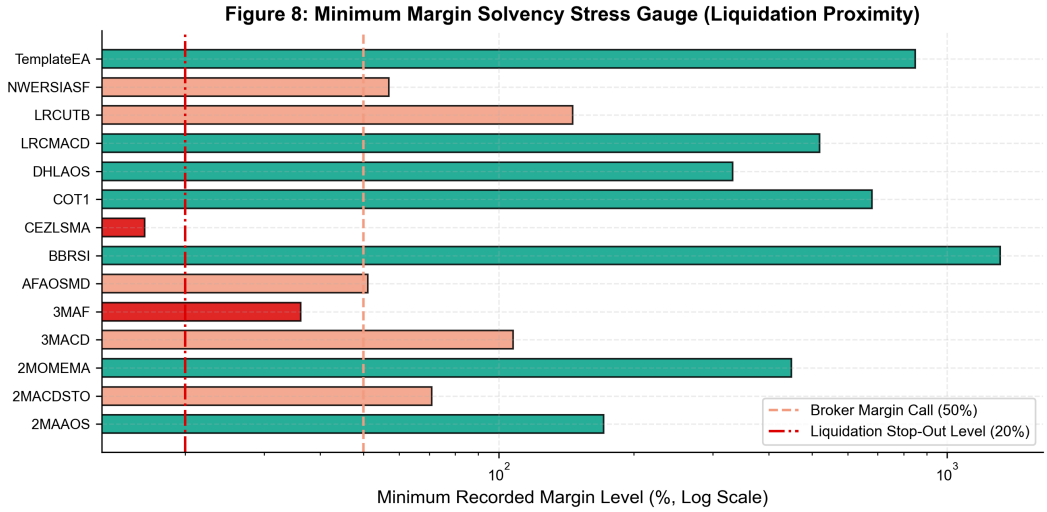


Figure 8: Minimum Recorded Margin Level Stress Gauge. Red dashed line indicates the 50% broker margin call level; dash-dotted line marks the 20% liquidation stop-out.

Figure 9 evaluates the duration-PnL asymmetry across all recorded trades, confirming that grid strategies hold losing positions for extended time horizons.

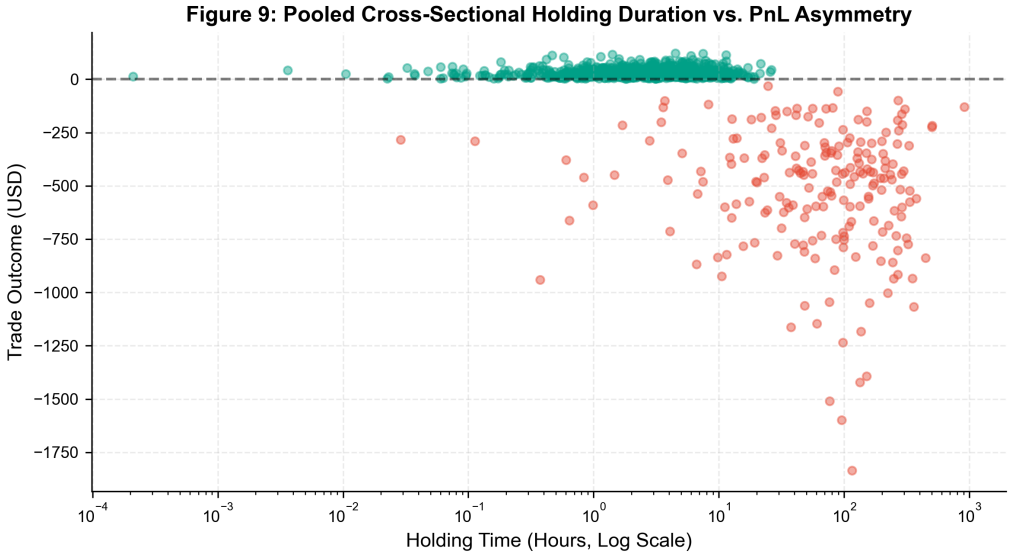


Figure 9: Pooled Holding Duration versus Trade Outcome. Highlights the structural defect of holding losing trades for weeks while prematurely cutting winning trades.

Figure 10 presents the 1,000-path Monte Carlo bootstrap simulation, illustrating the widening Value-at-Risk envelope over a 250-day forward horizon.

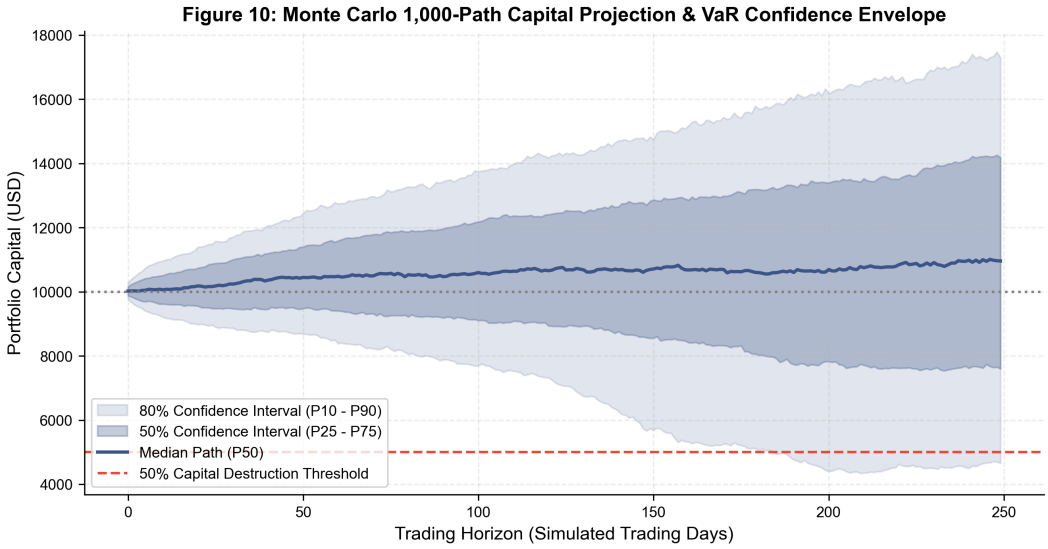


Figure 10: Monte Carlo 1,000-Path Capital Projection and Confidence Envelope.

7 The Berrytrade Governance Framework

To protect open-source quantitative practitioners from catastrophic risk, the **Berrytrade-EA** ecosystem institutes an automated two-tier funnel:

1. **Experimental Tier (Incubation):** Allows community contributions for algorithmic exploration, but mandates strict metadata declaration (`uses_grid: true`, `risk_level: "Extreme"`).

2. **Verified Tier (Production-Grade):** Imposes non-negotiable hurdles: 3+ years tick data, maximum equity drawdown $\leq 20\%$, profit factor ≥ 1.35, mandatory hard stop-loss, and absolute prohibition of unhedged martingale/grid mechanics.

3. **Automated Continuous Integration:** GitHub Actions pipelines scan every PR for forbidden native Win32 DLL calls (`user32.dll`, `kernel32.dll`) and undeclared telemetry, ensuring complete code transparency.

8 Conclusion

This study has provided a rigorous empirical evaluation of retail Expert Advisors, uncovering the systematic substitution of tail risk for true statistical alpha. The popular practice of superimposing grid overlays onto unprofitable indicator signals creates a deceptive facade of steady gains that inevitably terminates in account liquidation during sustained trending regimes.

Through open-source dissemination, standardized benchmarking, and transparent peer review, the **Berrytrade-EA** project aims to elevate retail algorithmic trading to institutional standards of scientific reproducibility, risk discipline, and capital preservation.

References

[1] David H Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

[2] Eugene F Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970.

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